

D907

Health Services Research

Armored Drill

8 concepts · 12 items · 4 transfer items

What this is. Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

Source. Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

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How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	_____	

Items

1.

A researcher at a university with a renowned medical library studies why rural adults delay colon cancer screening. She finds that many lack health insurance and live more than an hour from a clinic. Her team won a large federal grant last year. In the Andersen model, insurance and travel distance are which factors?

- A. Predisposing factors
- B. Need factors
- C. Enabling factors
- D. Realized access, since they explain use

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

2.

A drug lowered blood pressure by 15 mmHg in a tightly controlled trial of carefully selected patients at an academic center with a new research building. When a health system rolled it out, patients who sometimes miss doses averaged a 6 mmHg drop. The system's CEO is new. The 6 mmHg result describes the drug's:

- A. Proof that the trial was invalid
- B. Efficacy
- C. Safety profile
- D. Effectiveness

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

3.

A state report ranks two hospitals by raw 30-day mortality: Hospital A at 6% and Hospital B at 3%. A is a regional trauma and transplant center that receives the sickest transfers. B is a suburban community hospital with a new parking garage. What should an analyst do before concluding A has worse quality?

- A. Use raw rates, because adjustment always hides poor care
- B. Risk-adjust the mortality rates for patient severity and case mix
- C. Nothing; A is worse because 6% is double 3%
- D. Adjust for amenities such as parking and room quality

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

4.

A trial at a single academic center, known for its winning basketball team, randomized carefully screened patients under 60 with no other illnesses and found a strong effect. The trial had excellent follow-up. A rural system serving mostly older adults with several chronic conditions asks whether the result applies to its patients. What is the main concern?

- A. Bias from the center's basketball publicity
- B. Internal validity
- C. Confounding from failed randomization
- D. External validity

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

5.

Researchers at a health department housed in a historic courthouse identify 200 people who developed a rare liver disease and 400 similar people who did not. They then look back at each person's past exposure to an industrial solvent. The team leader speaks three languages. What design is this?

- A. Randomized controlled trial
- B. Case-control study
- C. Cross-sectional study
- D. Prospective cohort study

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

6. [TRANSFER — reversed stem / different variable]

Researchers enroll 5,000 nurses without heart disease, record their shift-work patterns, and follow them for 10 years to see who develops heart disease. What design is this?

- A. Prospective cohort study
- B. Case-control study
- C. Randomized controlled trial
- D. Cross-sectional study

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

7.

In a trial across a clinic network that recently adopted a new EHR, a fall-prevention program lowered one-year hip-fracture rates from 4% to 3% among older adults. The program is funded by a local foundation. What is the absolute risk reduction?

- A. 25%
- B. 3%
- C. 1 percentage point
- D. 7%

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

8. [TRANSFER — reversed stem / different variable]

Using the same result, a drop in fracture risk from 4% to 3%, how many older adults must receive the program to prevent one hip fracture?

- A. 33
- B. 0.01
- C. 100
- D. 4

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

9.

A researcher at an agency whose director recently testified before Congress wants survey results that represent all Medicare enrollees in her state. She has a list of every enrollee. Which step best supports generalizing the results?

- A. Draw a random sample from the full enrollee list
- B. Randomly assign respondents to two groups
- C. Survey people who respond to news coverage of the director's testimony
- D. Survey volunteers at one large clinic

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

10. [TRANSFER — reversed stem / different variable]

A researcher wants to show that a care-coordination program causes fewer readmissions. Which step most strengthens the causal claim?

- A.** Randomly assign eligible patients to the program or usual care
- B.** Randomly sample patients from the hospital's discharge list
- C.** Compare patients who volunteered with those who declined
- D.** Enroll as many patients as possible, since a big sample always proves cause

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

11.

State X, which has a thriving tech sector, expanded Medicaid; State Y did not. Over the same years, X's uninsured rate fell from 18% to 10% and Y's fell from 17% to 14%. Both states have about 5 million residents. What is the difference-in-differences estimate of the expansion's effect?

- A.** An 11-point reduction
- B.** An 8-point reduction
- C.** A 4-point reduction
- D.** A 5-percentage-point reduction

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

12. [TRANSFER — reversed stem / different variable]

A difference-in-differences study found a program cut ED visits by 12 per 1,000 more than its comparison area. The comparison area fell from 80 to 74 per 1,000. The program area started at 90 per 1,000. Where did the program area end?

- A.** 96 per 1,000
- B.** 72 per 1,000
- C.** 78 per 1,000
- D.** 84 per 1,000

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

STOP

Answers and dissection follow. Finish every item and every strip first.

Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

1. Answer C Concept 1

A researcher at a university ~~with a renowned medical library~~ studies why rural adults delay colon cancer screening. She finds that many lack health insurance and live more than an hour from a clinic. ~~Her team won a large federal grant last year.~~ In the Andersen model, insurance and travel distance are which factors?

Target word: Andersen factor type **Decides:** Insurance and distance are resources and logistics.

A	Confusion set	Predisposing factors are demographics and beliefs.
B	Adjacent category	Need factors are perceived or measured health status.
C	Correct	Insurance, income, and distance enable or block use.
D	Inverted operation	Realized access is the use itself, the outcome, not the determinants.

2. Answer D Concept 2

A drug lowered blood pressure by 15 mmHg in a tightly controlled trial of carefully selected patients at an academic center ~~with a new research building~~. When a health system rolled it out, patients who sometimes miss doses averaged a 6 mmHg drop. ~~The system's CEO is new.~~ The 6 mmHg result describes the drug's:

Target word: efficacy or effectiveness **Decides:** Real-world routine use.

A	Absolute language	A smaller real-world effect is expected, not evidence of a bad trial.
B	Confusion set	Efficacy is the 15 mmHg ideal-condition result.
C	Adjacent category	Safety concerns harms, not the size of the benefit.
D	Correct	Results under routine, real-world conditions measure effectiveness.

3. Answer B Concept 3

A state report ranks two hospitals by raw 30-day mortality: Hospital A at 6% and Hospital B at 3%. A is a regional trauma and transplant center that receives the sickest transfers. B is a suburban community hospital ~~with a new parking garage~~. What should an analyst do before concluding A has worse quality?

Target word: step before comparing **Decides:** A treats much sicker patients.

A	Absolute language	Adjustment makes comparisons fair; it does not always hide problems.
B	Correct	Raw rates mix quality with how sick the patients were.
C	Dropped element	Drops the case-mix difference that the comparison depends on.
D	Camouflage	Amenities do not explain patient severity.

4. Answer D Concept 4

A trial at a single academic center, ~~known for its winning basketball team~~, randomized carefully screened patients under 60 with no other illnesses and found a strong effect. ~~The trial had excellent follow-up.~~ A rural system serving mostly older adults with several chronic conditions asks whether the result applies to its patients. What is the main concern?

Target word: which validity **Decides:** Whether results transfer to a different population.

A	Camouflage	The team is irrelevant to the study.
B	Confusion set	Randomization and good follow-up support internal validity.
C	Adjacent category	Nothing suggests randomization failed.
D	Correct	The question is whether the result generalizes to older, sicker patients.

5. Answer B Concept 5

Researchers at a health department ~~housed in a historic courthouse~~ identify 200 people who developed a rare liver disease and 400 similar people who did not. They then look back at each person's past exposure to an industrial solvent. ~~The team leader speaks three languages.~~ What design is this?

Target word: study design **Decides:** Selected by disease status, then looked back.

A	Adjacent category	No exposure was assigned.
B	Correct	Starting from the outcome and looking back at exposure is case-control.
C	Confusion set	Cross-sectional measures exposure and outcome at one time.
D	Inverted operation	A cohort starts from exposure and moves forward.

6. Answer A Concept 5 • transfer

Researchers enroll 5,000 nurses without heart disease, record their shift-work patterns, and follow them for 10 years to see who develops heart disease. What design is this?

Target word: study design **Decides:** Selected by exposure, followed forward.

A	Correct	Starting with exposure and following forward is a cohort.
B	Inverted operation	That starts from the disease and looks back.
C	Adjacent category	Shift work was observed, not assigned.
D	Confusion set	That uses a single time point, not ten years of follow-up.

7. Answer C Concept 6

In a trial across a clinic network ~~that recently adopted a new EHR~~, a fall-prevention program lowered one-year hip-fracture rates from 4% to 3% among older adults. ~~The program is funded by a local foundation.~~ What is the absolute risk reduction?

Target word: ARR **Decides:** $4\% - 3\%$.

A	Confusion set	That is the relative risk reduction ($1 \div 4$).
B	Wrong stem number	That is the treatment-group rate.
C	Correct	$ARR = 4\% - 3\% = 1$ percentage point.
D	Inverted operation	Adds the rates instead of subtracting.

8. Answer C Concept 6 • transfer

Using the same result, a drop in fracture risk from 4% to 3%, how many older adults must receive the program to prevent one hip fracture?

Target word: NNT **Decides:** $1 \div 0.01$.

A	Wrong stem number	Divides by the 3% treatment rate.
B	Inverted operation	Reports the ARR without inverting it.
C	Correct	$NNT = 1 \div ARR = 1 \div 0.01 = 100$.
D	Confusion set	Uses the 25% relative reduction instead of the absolute one.

9. Answer A Concept 7

A researcher at an agency ~~whose director recently testified before Congress~~ wants survey results that represent all Medicare enrollees in her state. She has a list of every enrollee. Which step best supports generalizing the results?

Target word: step for generalizing **Decides:** Representativeness comes from random sampling.

A	Correct	Random sampling makes the sample representative of the population.
B	Confusion set	Random assignment supports causal claims, not representativeness.
C	Camouflage	The testimony is irrelevant, and self-selected responders are not representative.
D	Adjacent category	A convenience sample may not represent the state.

10. Answer A Concept 7 · transfer

A researcher wants to show that a care-coordination program causes fewer readmissions. Which step most strengthens the causal claim?

Target word: step for causation **Decides:** Random assignment balances groups.

A	Correct	Random assignment balances known and unknown confounders.
B	Confusion set	Sampling supports generalizing, not cause.
C	Adjacent category	Volunteers differ in ways that bias the comparison.
D	Absolute language	Size improves precision, not causal inference.

11. Answer D Concept 8

State X, ~~which has a thriving tech sector~~, expanded Medicaid; State Y did not. Over the same years, X's uninsured rate fell from 18% to 10% and Y's fell from 17% to 14%. ~~Both states have about 5 million residents.~~ What is the difference-in-differences estimate of the expansion's effect?

Target word: DiD estimate **Decides:** $(-8) - (-3)$.

A	Inverted operation	Adds the two changes instead of subtracting.
B	Dropped element	Drops the comparison state's change.
C	Confusion set	Compares only the final rates $(14\% - 10\%)$.
D	Correct	X changed -8 points, Y changed -3 points; $-8 - (-3) = -5$.

12. Answer B Concept 8 · transfer

A difference-in-differences study found a program cut ED visits by 12 per 1,000 more than its comparison area. The comparison area fell from 80 to 74 per 1,000. The program area started at 90 per 1,000. Where did the program area end?

Target word: program-area end value **Decides:** Program change = $-6 - 12 = -18$.

A	Inverted operation	Adds the comparison change instead of subtracting.
B	Correct	Program change = comparison change (-6) + effect $(-12) = -18$; $90 - 18 = 72$.
C	Dropped element	Drops the comparison area's change.
D	Confusion set	Applies only the comparison area's change.

Triage grid

Write each item number in the box it belongs to.

	Right	Wrong
Sure	Owned.	Concept inversion — top priority. A wrong mental model; it survives rewording.
Unsure / guess	Recognition, not mastery. Re-drill with the transfer.	Not learned yet.

Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

Trap	What it does	In this drill	Your misses
Dropped element	Leaves out a required term or step.	3	
Wrong stem number	A figure copied from the prompt that is not the answer.	2	
Adjacent category	The right idea in the wrong bucket.	7	
Inverted operation	Direction flipped: add/subtract, cause/effect, higher/lower.	7	
Absolute language	Always, never, guarantee — where judgment applies.	3	
Confusion set	Every option is real; the test is exact role-matching.	11	
Camouflage	Answers a stem fact that was never the question.	3	

Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

#	Concept	Rule	Trap	Tell
1	Andersen behavioral model of access	Use of services is shaped by predisposing factors (age, beliefs, education), enabling factors (insurance, income, distance), and need (perceived or evaluated illness).	Labeling insurance or distance as predisposing, or confusing determinants with actual use.	Resources and logistics versus demographics and beliefs versus health status.
2	Efficacy vs. effectiveness	Efficacy is the effect under ideal, controlled conditions; effectiveness is the effect in routine real-world practice.	Treating a trial result as what a health system will see.	Tightly selected trial patients versus a real-world rollout.
3	Risk adjustment	Compare outcomes across providers only after adjusting for patient severity and case mix.	Ranking hospitals on raw mortality or readmission rates.	A hospital that receives the sickest or most complex patients.
4	Internal vs. external validity	Internal validity is whether the study's result is true for its sample; external validity is whether it generalizes to other people and settings.	Assuming a well-run trial automatically applies to every population.	"Do these results apply to our patients?"
5	Cohort vs. case-control design	A cohort starts from exposure and follows people forward to outcomes; a case-control study starts from the outcome and looks back at past exposure.	Calling any study that looks at exposure a cohort study.	Whether participants were selected by disease status or by exposure.
6	Absolute vs. relative risk and NNT	Absolute risk reduction = control rate – treatment rate; relative risk reduction = $ARR \div \text{control rate}$; $NNT = 1 \div ARR$.	Presenting the relative reduction as the absolute one.	Two event rates and a question about "how many need to be treated."
7	Random sampling vs. random assignment	Random sampling supports generalizing to a population; random assignment supports causal claims by balancing groups.	Thinking random sampling proves cause or random assignment makes results representative.	"Represent the population" versus "show the program caused it."
8	Difference-in-differences	The policy effect = (change in the treated group) – (change in the comparison group).	Crediting the policy with the whole before-after change in the treated group.	A policy group and a comparison group, each with before and after values.

Course scope (public description): Health Services Research provides an in-depth introduction to the field of healthcare research focusing on the systematic investigation of how healthcare services are accessed, the quality of care provided, and the outcomes of care. It is designed to equip learners with the knowledge and skills needed to critically evaluate healthcare systems, policies, and practices that inform health services delivery and policy decisions. Prerequisite: There are no prerequisites to the course.